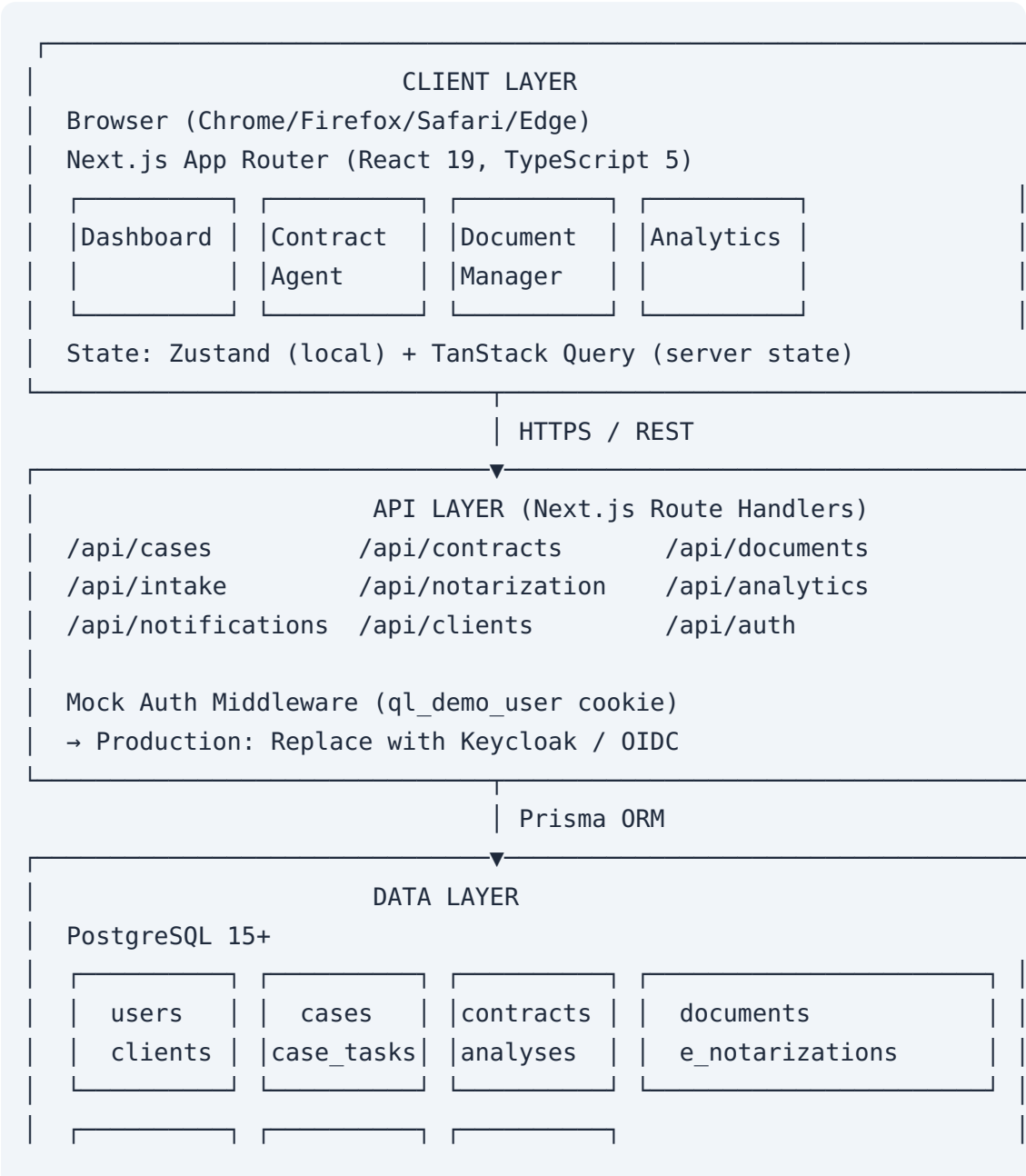
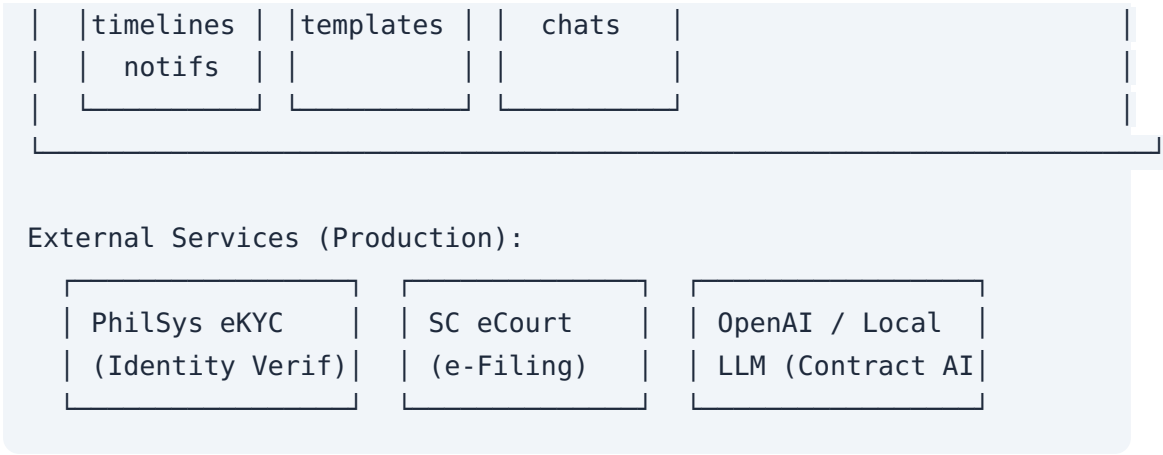


Quanby Case Management Platform — Technical Overview

Version: 1.0 | **Classification:** Internal / Technical
Standards: NIST SP 800-53 Rev. 5, ISO/IEC 27001:2022, OWASP Top 10
Applicable Laws: RA 10173, RA 8792

1. Architecture Diagram





2. Tech Stack Summary

Layer	Technology	Version	Purpose
Framework	Next.js	15.1.6	Full-stack React framework (App Router)
Language	TypeScript	5.7.x	Type safety throughout
Runtime	Node.js	18+	Server runtime
UI Library	React	19.0.0	Component rendering
Styling	Tailwind CSS	3.4.x	Utility-first CSS
Components	shadcn/ui + Radix UI	Latest	Accessible UI primitives
Animation	Framer Motion	12.x	Micro-interactions
Icons	Lucide React	0.474	Icon system
State (server)	TanStack Query	5.65	API data fetching + caching
State (client)	Zustand	5.0	Global UI state
Forms	React Hook Form + Zod	7.x + 3.x	Form management + validation
ORM	Prisma	6.3.0	Type-safe DB access
Database	PostgreSQL	15+	Primary data store
PDF Generation	@react-pdf/renderer	4.x	In-browser PDF generation

Layer	Technology	Version	Purpose
Charts	Recharts	2.15	Analytics visualizations
Notifications	Sonner	1.7	Toast notifications
Themes	next-themes	0.4	Dark/light mode
Date Handling	date-fns	4.x	Date formatting and calculations

3. Database Schema Overview

Core Models

users	Primary identity record
├ clerkId (String)	Unique auth provider ID (mock in demo)
├ email, firstName, lastName	
├ role (ADMIN LAWYER PARALEGAL CLIENT)	
├ barNumber, specializations[]	(Lawyers only)
└ maxCaseLoad (Int)	
clients	Client personal + eligibility data
├ userId (→ users)	
├ Personal info (DOB, gender, civil status, nationality)	
├ Address (street, barangay, municipality, province, region)	
├ Government IDs (TIN, SSS, PhilHealth, Pag-IBIG, etc.)	
├ Financial (monthlyIncome, occupation, employer)	
└ Eligibility (status, score, notes, reviewedBy)	
cases	Core legal case record
├ caseNumber (String @unique)	
├ type (CaseType enum: CIVIL CRIMINAL LABOR ...)	
├ status (CaseStatus enum: INTAKE→...→ARCHIVED)	
├ priority (CasePriority: URGENT HIGH MEDIUM LOW)	
├ Court info (courtLevel, courtName, courtBranch, docket, judge)	
├ Dates (filing, incident, reglementaryDeadline, nextHearing)	
├ clientId (→ clients)	
└ assignedLawyerId (→ users)	
case_timelines	Immutable audit trail per case
case_tasks	To-do items per case
documents	All legal documents

- └─ type (DocumentType enum: COMPLAINT|ANSWER|MOTION|...)
- └─ status (DRAFT→REVIEW→APPROVED→FILED→SERVED→ARCHIVED)
- └─ File metadata (fileName, fileSize, mimeType, storageKey)
- └─ Generated document fields (templateId, generatedContent)
- └─ Notarization fields (isNotarized, notaryName, etc.)
- └─ eNotarization (→ e_notarizations)

contracts ————— Contract Agent documents

- └─ type (ContractType enum: NDA|SERVICE|EMPLOYMENT|...)
- └─ status (ContractStatus enum: UPLOADED→ANALYZING→ANALYZED→...)
- └─ parties (Json array: [{name, role, address}])
- └─ value, currency
- └─ linkedCaseId (→ cases) optional case linkage

contract_analyses ————— AI analysis results per contract

- └─ overallRiskScore (Int 0-100)
- └─ riskLevel (LOW|MEDIUM|HIGH|CRITICAL)
- └─ keyClauses (Json)
- └─ risks (Json)
- └─ complianceFlags (Json)
- └─ suggestions (Json)
- └─ scCompliant (Boolean)

contract_templates ————— Reusable contract templates

contract_chats ————— Per-contract AI chat history

e_notarizations ————— Notarization records

- └─ certificateNumber (String @unique)
- └─ status (PENDING|VERIFIED|NOTARIZED|REJECTED)
- └─ Notary details (name, IBP roll, PTR, MCLE)
- └─ verificationHash (SHA-256)
- └─ verificationUrl

notifications ————— User notification queue

Enums Reference

Role:	ADMIN LAWYER PARALEGAL CLIENT
CaseType:	CIVIL CRIMINAL LABOR ADMINISTRATIVE FAMILY COMMERCIAL SPECIAL_PROCEEDINGS ELECTION TAX ENVIRONMENTAL
CaseStatus:	INTAKE SCREENING ASSIGNED PENDING_ASSIGNMENT ACTIVE ON_HOLD HEARING AWAITING_HEARING SUBMITTED AWAITING_DECISION DECIDED APPEALED CLOSED CLOSED_WON CLOSED_LOST CLOSED_SETTLED DISMISSED ARCHIVED

CasePriority: URGENT | HIGH | MEDIUM | LOW
CourtLevel: BARANGAY | MTC | MTCC | RTC | CA | SB | SC | NLRC |
DARAB | COMELEC | CTA | OMBUDSMAN
ContractType: EMPLOYMENT | SERVICE | LEASE | SALE | PARTNERSHIP | LOAN |
NDA | MOA | MOU | DEED_OF_SALE | DEED_OF_DONATION |
POWER_OF_ATTORNEY | FRANCHISE | CONSTRUCTION | SUPPLY | OTHER

4. API Endpoint Reference

Authentication

Method	Endpoint	Description
POST	/api/auth/set-user	Set demo user cookie ({userId: string})

Cases

Method	Endpoint	Description
GET	/api/cases	List cases (query: status , type , lawyerId , page , limit)
POST	/api/cases	Create new case
GET	/api/cases/[id]	Get case by ID (includes client, lawyer, recent timeline)
PATCH	/api/cases/[id]	Update case (status, priority, court info, etc.)
DELETE	/api/cases/[id]	Archive/delete case
GET	/api/cases/[id]/timeline	Get case timeline events
POST	/api/cases/[id]/timeline	Add timeline event
GET	/api/cases/[id]/tasks	List case tasks
POST	/api/cases/[id]/tasks	Create case task

Clients

Method	Endpoint	Description
GET	/api/clients	List clients
POST	/api/clients	Create client record

Client Intake

Method	Endpoint	Description
POST	/api/intake	Submit intake form (creates client + case)

Documents

Method	Endpoint	Description
GET	/api/documents	List documents (query: <code>caseId</code> , <code>type</code> , <code>status</code>)
POST	/api/documents	Upload document metadata
GET	/api/documents/[id]	Get document by ID
PATCH	/api/documents/[id]	Update document status/metadata
DELETE	/api/documents/[id]	Delete document
POST	/api/documents/generate	Generate document from template

Contracts

Method	Endpoint	Description
GET	/api/contracts	List contracts
POST	/api/contracts	Create/upload contract
GET	/api/contracts/[id]	Get contract with analyses and chats
PATCH	/api/contracts/[id]	Update contract
DELETE	/api/contracts/[id]	Delete contract

Method	Endpoint	Description
POST	/api/contracts/analyze	Trigger AI analysis for a contract
GET	/api/contracts/[id]/chat	Get chat history
POST	/api/contracts/[id]/chat	Send chat message to Contract Agent
GET	/api/contracts/templates	List contract templates

Notarization

Method	Endpoint	Description
GET	/api/notarization	List notarization requests
POST	/api/notarization	Create notarization request
PATCH	/api/notarization	Update notarization status (verify/notarize/reject)

Analytics

Method	Endpoint	Description
GET	/api/analytics	Get analytics data (query: range , lawyerId)

Notifications

Method	Endpoint	Description
GET	/api/notifications	List notifications for current user
PATCH	/api/notifications	Mark notification(s) as read

5. Authentication System

Demo Mode (Current Implementation)

The platform uses a **cookie-based mock authentication** system for demo purposes.

Cookie name: ql_demo_user

Cookie type: httpOnly , sameSite: lax , maxAge: 7 days

Secure flag: Enabled in production environment

User Resolution Priority: 1. `?user=<id>` query parameter (overrides cookie, sets cookie) 2. `ql_demo_user` cookie 3. Fallback: User ID "1" (Atty. Maria Santos, Admin)

Key files: - `src/lib/auth.ts` — Server-side auth utilities (`requireAuth`, `getCurrentUser`) - `src/lib/auth-client.ts` — Client-safe constants (`DEMO_USERS`, `MOCK_USER_COOKIE`)

Note: The `clerkId` field in the database schema is a legacy column name preserved for migration compatibility. It is populated with synthetic IDs (`system-intake-*` , `manual-*`) and does not require Clerk in demo mode.

Production Migration Path

Replace mock auth with:

```
Keycloak (recommended for government)
→ OIDC/OAuth2 flow
→ TOTP (RA 10173 requirement for sensitive systems)
→ Replace getCurrentUser() with Keycloak session validation
→ Map Keycloak user ID to users.clerkId column
→ Implement proper RBAC middleware
```

6. Deployment Guide

Prerequisites

```
Node.js 18+
PostgreSQL 15+
npm or pnpm
```

Step 1: Clone and Install

```
git clone <repository-url>
cd quanby-legal-demo
npm install
```

Step 2: Configure Environment

Copy `.env.example` to `.env.local`:


```
cp .env.example .env.local
```

Edit `.env.local` with your values (see Environment Variables section below).

Step 3: Database Setup

```
# Create database
createdb quanby_legal_dev

# Run migrations
npm run db:migrate

# Generate Prisma client
npm run db:generate

# Seed demo data
npm run db:seed
```

Step 4: Development Server

```
npm run dev
# Open: http://localhost:3000
```

Step 5: Production Build

```
# Build
npm run build

# Type check (optional, recommended before deploy)
npm run type-check

# Start production server
npm start
```

Docker Deployment (Optional)

```
FROM node:18-alpine
WORKDIR /app
COPY package*.json ./
RUN npm ci --production
COPY . .
```

```
RUN npm run build
EXPOSE 3000
CMD ["npm", "start"]
```

```
# docker-compose.yml
version: '3.8'
services:
  app:
    build: .
    ports:
      - "3000:3000"
    environment:
      DATABASE_URL: postgresql://postgres:password@db:5432/quantby_legal
    depends_on:
      - db
  db:
    image: postgres:15-alpine
    environment:
      POSTGRES_DB: quantby_legal
      POSTGRES_PASSWORD: password
    volumes:
      - postgres_data:/var/lib/postgresql/data
volumes:
  postgres_data:
```

7. Environment Variables Reference

Variable	Required	Description	Example
<code>DATABASE_URL</code>	☐	PostgreSQL connection string	<code>postgresql://user:pass@localhost:5432/quantby_legal</code>
<code>NEXTAUTH_SECRET</code>	Production	Session secret (32+ chars)	<code>openssl rand -base64 32</code>
<code>NEXTAUTH_URL</code>	Production	Canonical app URL	<code>https://legal.quantby.com</code>
<code>NEXT_PUBLIC_APP_URL</code>	Optional	Public-facing URL	<code>https://legal.quantby.com</code>

Variable	Required	Description	Example
<code>OPENAI_API_KEY</code>	Optional	For real AI contract analysis	<code>sk-...</code>
<code>ANTHROPIC_API_KEY</code>	Optional	Alternative LLM provider	<code>sk-ant-...</code>
<code>STORAGE_BUCKET</code>	Optional	S3-compatible bucket for files	<code>quanby-legal-docs</code>
<code>STORAGE_ENDPOINT</code>	Optional	S3 endpoint URL	<code>https://s3.amazonaws.com</code>
<code>STORAGE_ACCESS_KEY</code>	Optional	S3 access key	<code>AKIA...</code>
<code>STORAGE_SECRET_KEY</code>	Optional	S3 secret key	<code>...</code>
<code>SMTP_HOST</code>	Optional	Email server for notifications	<code>smtp.mailgun.org</code>
<code>SMTP_PORT</code>	Optional	SMTP port	<code>587</code>
<code>SMTP_USER</code>	Optional	SMTP username	<code>postmaster@quanby.com</code>
<code>SMTP_PASS</code>	Optional	SMTP password	<code>...</code>
<code>NODE_ENV</code>	Auto	Environment (development/production)	Set by runtime

8. Known Limitations (Demo Mode)

The following features are **simulated** or **mocked** in the current demo version:

Feature	Current State	Production Implementation
AI Contract Analysis	Returns pre-seeded mock analysis data	Integrate OpenAI GPT-4 / Anthropic Claude with Philippine law system prompt
Contract Chat	Returns scripted mock responses	Stream responses from LLM API with contract context injection
File Upload	Metadata stored; no actual file storage	Integrate S3-compatible storage (AWS S3, MinIO, or local NAS)
Identity Verification	Step UI shown; no real eKYC	Integrate PhilSys eKYC API or Jumio/ Onfido
Digital Signatures	Certificate generated; no real PKI	Integrate with PGP or certificate authority
Email Notifications	Notification records created in DB; no email sent	Configure SMTP + email templates
SMS Notifications	Not implemented	Integrate Globe/Smart SMS API or Twilio
Court e-Filing	Not implemented	Integrate SC eCourt API when available
Payment/Billing	Not implemented	Integrate DragonPay, PayMaya, or GCash
Real-time Updates	Page refresh required	Implement WebSocket or SSE for live updates
Multi-tenancy	Single-firm only	Add <code>firmId</code> field to all models for multi-firm
Audit Logging	Application-level only	Add database audit triggers for RA 10173 compliance
Encryption at Rest	Not implemented	Enable PostgreSQL transparent data encryption
2FA / MFA	Not implemented	Required for production under NIST SP 800-53 IA-5

9. Security Notes (NIST SP 800-53 / ISO 27001)

Implemented (Demo)

- Role-based access control at API level
- HttpOnly, Secure, SameSite cookies
- Input validation via Zod schemas
- SQL injection prevention via Prisma parameterized queries
- CSRF protection via Next.js default headers

Required Before Production

Control	Standard	Action Required
Multi-factor authentication	NIST IA-5	Implement TOTP (Google Authenticator, Authy)
Data encryption at rest	NIST SC-28	Enable PostgreSQL TDE or column-level encryption
TLS 1.3	NIST SC-8	Configure via reverse proxy (nginx/Caddy)
Security headers	OWASP	Add CSP, HSTS, X-Frame-Options headers
Vulnerability scanning	NIST RA-5	Integrate Snyk or OWASP Dependency-Check in CI/CD
Penetration testing	ISO 27001 A.12.6	Annual pentest by accredited firm
Data backup	ISO 27001 A.12.3	Daily encrypted backups to offsite storage
Incident response plan	NIST IR-1	Document and test IR procedures
Privacy Impact Assessment	RA 10173 §76	Required before go-live
NPC registration	RA 10173 §46	Register as Personal Information Controller

10. Project Structure

```
quanby-legal-demo/  
├─ prisma/
```

```

|   ├── schema.prisma          # Database schema
|   └── seed.ts                # Demo data seeder
├── public/
|   ├── logo.svg              # Quanby Legal logo
|   └── sc-badge.svg          # Supreme Court compliance badge
├── src/
|   ├── app/
|   |   ├── (auth)/           # Login / signup pages
|   |   ├── (dashboard)/      # Protected dashboard pages
|   |   |   ├── layout.tsx    # Dashboard shell (sidebar + topnav)
|   |   |   ├── dashboard/    # Main dashboard
|   |   |   ├── cases/        # Case list + detail + new
|   |   |   ├── intake/       # Client intake wizard
|   |   |   ├── contracts/    # Contract Agent
|   |   |   ├── documents/    # Document management
|   |   |   ├── notarization/ # e-Notarization
|   |   |   ├── portal/       # Client portal
|   |   |   ├── analytics/    # Analytics dashboard
|   |   |   └── settings/     # Settings page
|   |   ├── api/              # REST API route handlers
|   |   ├── globals.css       # Global styles + CSS variables
|   |   └── layout.tsx        # Root layout
|   ├── components/
|   |   ├── analytics/        # Chart components
|   |   ├── cases/            # Case-related components
|   |   ├── contracts/        # Contract Agent components
|   |   ├── documents/        # Document management components
|   |   ├── layout/           # Sidebar, TopNav, ComplianceBadge
|   |   ├── notifications/    # Notification panel
|   |   ├── portal/           # Client portal components
|   |   ├── providers/        # React Query provider
|   |   └── ui/                # shadcn/ui base components
|   ├── lib/
|   |   ├── auth.ts           # Server-side auth utilities
|   |   ├── auth-client.ts    # Client-safe auth constants
|   |   ├── constants.ts      # Philippine legal constants
|   |   ├── prisma.ts         # Prisma client singleton
|   |   └── utils.ts          # Shared utilities (cn, etc.)
|   ├── store/
|   |   └── useStore.ts        # Zustand global store
|   └── types/
|       └── index.ts          # Shared TypeScript types
├── docs/
|   ├── USER_MANUAL.md
|   └── PRESENTATION_GUIDE.md

```

```
|   └─ TECHNICAL_OVERVIEW.md
|─ .env.example
|─ package.json
|─ tailwind.config.ts
└─ tsconfig.json
```

Maintained by: Quanby Case Management Platform Engineering Team

Classification: Internal Technical Documentation

Recent Enhancements (v1.0.0 Demo Build)

Animated Dashboard Counters

- Component: `src/components/ui/animated-counter.tsx`
- Uses `requestAnimationFrame` with ease-out cubic easing
- Counts from 0 to target value over 1 second on mount
- Wrapped in `src/components/dashboard/StatCards.tsx` (client component)

Global Search

- Debounced search (300ms) in TopNav search bar
- Queries three API endpoints in parallel: `/api/cases?search=`, `/api/contracts?search=`, `/api/documents?search=`
- Results grouped by type with hover tooltips showing full titles
- Keyboard support: Escape to close

Risk Gauge Animation

- SVG `stroke-dashoffset` transition on mount (1 second, cubic-bezier)
- Component: `src/components/contracts/RiskGauge.tsx`

QR Code on Notarization Certificates

- Decorative SVG QR pattern component: `src/components/documents/QRCode.tsx`
- Renders verification URL as visual QR pattern on certificate

Print/PDF Export (Analytics)

- `@media print` CSS block in `globals.css`
- Hides sidebar, navigation, buttons
- Formats content for A4 paper with proper margins

- Triggered via `window.print()` / Ctrl+P

Dark Mode Support

- Light/Dark toggle (System mode removed for simplicity)
- All dashboard stat cards have `dark:` Tailwind variants
- Theme persisted in localStorage via `next-themes`

Branding

- "Powered by Quanby AI" badge in dashboard footer
- Links to <https://quanbyai.com>